PROJECT | SOX + DATA ANALYTICS

I tested all 24,000 journal entries. Not a sample of 25.

Full-population journal entry testing with SQL for a simulated card-payments fintech.

198

high-risk manual entries flagged for follow-up

THE PROBLEM

Sampling can miss the one entry that matters

- Journal entries are a classic path for management override and error (PCAOB AS 2401 calls for testing them).
- Typical SOX tests pick 25–40 entries.
- Data tools now make it practical to test 100%, so every risky entry is reviewed, not just the lucky picks.

THE SETUP

NovaPay: a fictional card-payments fintech

24,000
FY2025 journal entry lines (synthetic)

5,293
manual entries

6
SQL risk tests

$50k
manual JE approval threshold

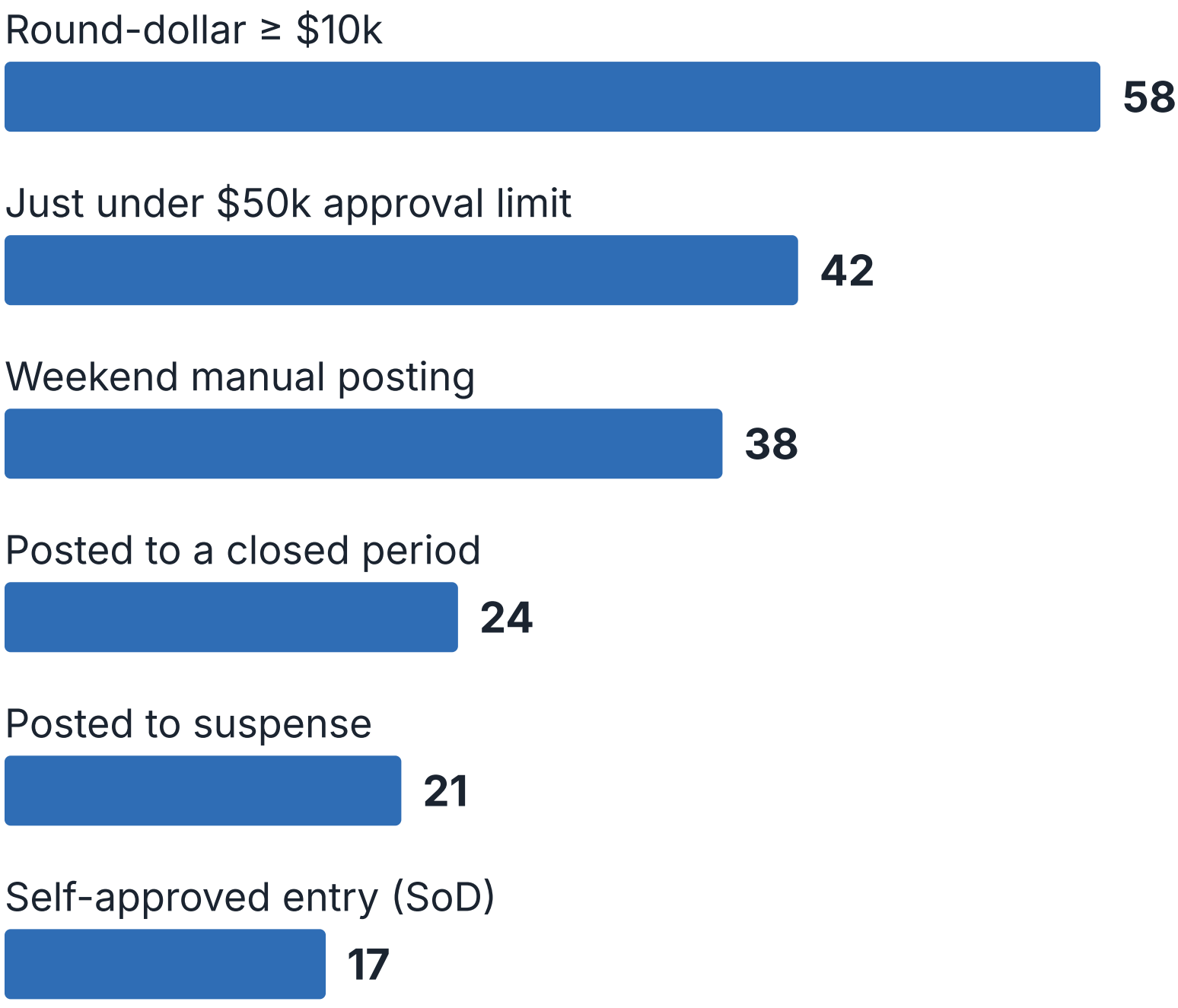
Built with Python (data), SQLite (tests) and Excel (results). Data is generated, so no real company information is used.

THE 6 TESTS

TEST	WHY IT MATTERS
Weekend posting	Outside normal review
Self-approved	Segregation of duties failure
Just under $50k	Splitting to avoid 2nd approval
Round-dollar ≥ $10k	Estimates or plugs
Suspense account	Unexplained balances
Closed period	Cut-off and close discipline

RESULTS

Entries flagged by test



198 unique entries = 3.7% of manual entries. 2 entries hit more than one test.

WHAT I WOULD DO NEXT

From exceptions to control findings

- **17 self-approved entries** point to a configuration gap in the ERP approval block: a design deficiency on the key JE control.
- **42 entries just under $50k**: review by preparer for splitting patterns.
- Feed the 6 tests into a monthly Controller review so the control is **detective every month**, not just at audit time.

TAKEAWAY

SOX expertise + SQL = better coverage with less sampling

10 years in audit, controls and SOX assurance taught me where the risk hides. Data lets me look at all of it.

Files (data, SQL, workbook) are linked in the post. Feedback welcome.